

The Question

We are analyzing a matrix (think of it as a table of numbers) that satisfies the following:

1. The roots of the **characteristic equation** of the matrix are all distinct (different) and real (not imaginary).
2. We need to check if the following assumptions are correct:
 - The matrix is **invertible**.
 - The matrix is **diagonalizable**.

We will explain all of this step by step, starting with the basic definitions.

Step 1: What is a Matrix?

- A **matrix** is just a rectangular array of numbers, like:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Here, A is a 2×2 matrix (2 rows and 2 columns).

Step 2: What is the Characteristic Equation?

1. The **characteristic equation** of a matrix is a special equation that helps us find the **eigenvalues** of the matrix. Eigenvalues are numbers that represent important properties of the matrix.
2. To compute the characteristic equation, we do the following:

- Subtract λ (a variable) from the diagonal entries of the matrix.
- Compute the determinant of the resulting matrix.

Example:

For the matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

the characteristic equation is found like this:

1. Subtract λ from the diagonal entries:

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & 2 \\ 3 & 4 - \lambda \end{bmatrix}.$$

2. Compute the determinant: The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is:

$$\text{Determinant} = ad - bc.$$

For $A - \lambda I$, the determinant is:

$$(1 - \lambda)(4 - \lambda) - (2)(3).$$

Simplify this:

$$\text{Determinant} = (1 - \lambda)(4 - \lambda) - 6 = \lambda^2 - 5\lambda - 2.$$

3. This gives us the **characteristic equation**:

$$\lambda^2 - 5\lambda - 2 = 0.$$

Step 3: Roots of the Characteristic Equation

1. The roots of the characteristic equation are the **eigenvalues** of the matrix. These are the solutions of the equation $\lambda^2 - 5\lambda - 2 = 0$.
 2. The problem tells us that the roots are **distinct** and **real**. For example, if the roots are $\lambda_1 = 1$ and $\lambda_2 = 4$, then these eigenvalues satisfy the given condition (they are real numbers, and they are different).
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Step 4: Assumption 1 – Is the Matrix Invertible?

1. A matrix is **invertible** if it has an inverse. Think of the inverse of a matrix like dividing numbers:
 - For a number a , the inverse is $\frac{1}{a}$.
 - For a matrix A , the inverse is another matrix A^{-1} , such that:

$$A \cdot A^{-1} = I,$$

where I is the **identity matrix**:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

2. How does invertibility relate to eigenvalues?

- A matrix is invertible if its determinant is NOT zero.
- The determinant of a matrix is the product of its eigenvalues:

$$\text{Determinant of } A = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n.$$
- If any eigenvalue is zero, the determinant becomes zero, and the matrix is NOT invertible.

3. Why is the matrix invertible here?

- The problem states that all eigenvalues are **distinct and real**. Since they are distinct, none of them can be zero.
- Thus, the determinant is nonzero, and the matrix is invertible.

Step 5: Assumption 2 – Is the Matrix Diagonalizable?

1. A matrix is **diagonalizable** if it can be written in the form:

$$A = PDP^{-1},$$

where:

- P is a matrix made up of the eigenvectors of A ,
 - D is a diagonal matrix containing the eigenvalues of A on its diagonal.
2. **When is a matrix diagonalizable?**
 - A matrix is diagonalizable if it has enough eigenvectors to form the matrix P .
 - If the eigenvalues are **distinct**, the matrix always has enough independent eigenvectors.
 3. **Why is the matrix diagonalizable here?**
 - The problem states that all eigenvalues are **distinct**. This guarantees that the matrix has n independent eigenvectors.
 - Therefore, the matrix is diagonalizable.

Step 6: Final Conclusion

The data scientist's assumptions are correct:

1. The matrix is **invertible** because all eigenvalues are nonzero (distinct roots mean no eigenvalue is zero).
 2. The matrix is **diagonalizable** because distinct eigenvalues guarantee that there are enough independent eigenvectors.
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Key Takeaway for a Beginner

- **Eigenvalues** are numbers that help us understand a matrix's properties. If they are distinct and real:
 - The matrix is **invertible** (if none of the eigenvalues are zero).
 - The matrix is **diagonalizable** (distinct eigenvalues guarantee enough eigenvectors).